

Differentiable Co-Optimization of Mass-Limited Legged Robots Across Variable Gravity



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Abstract

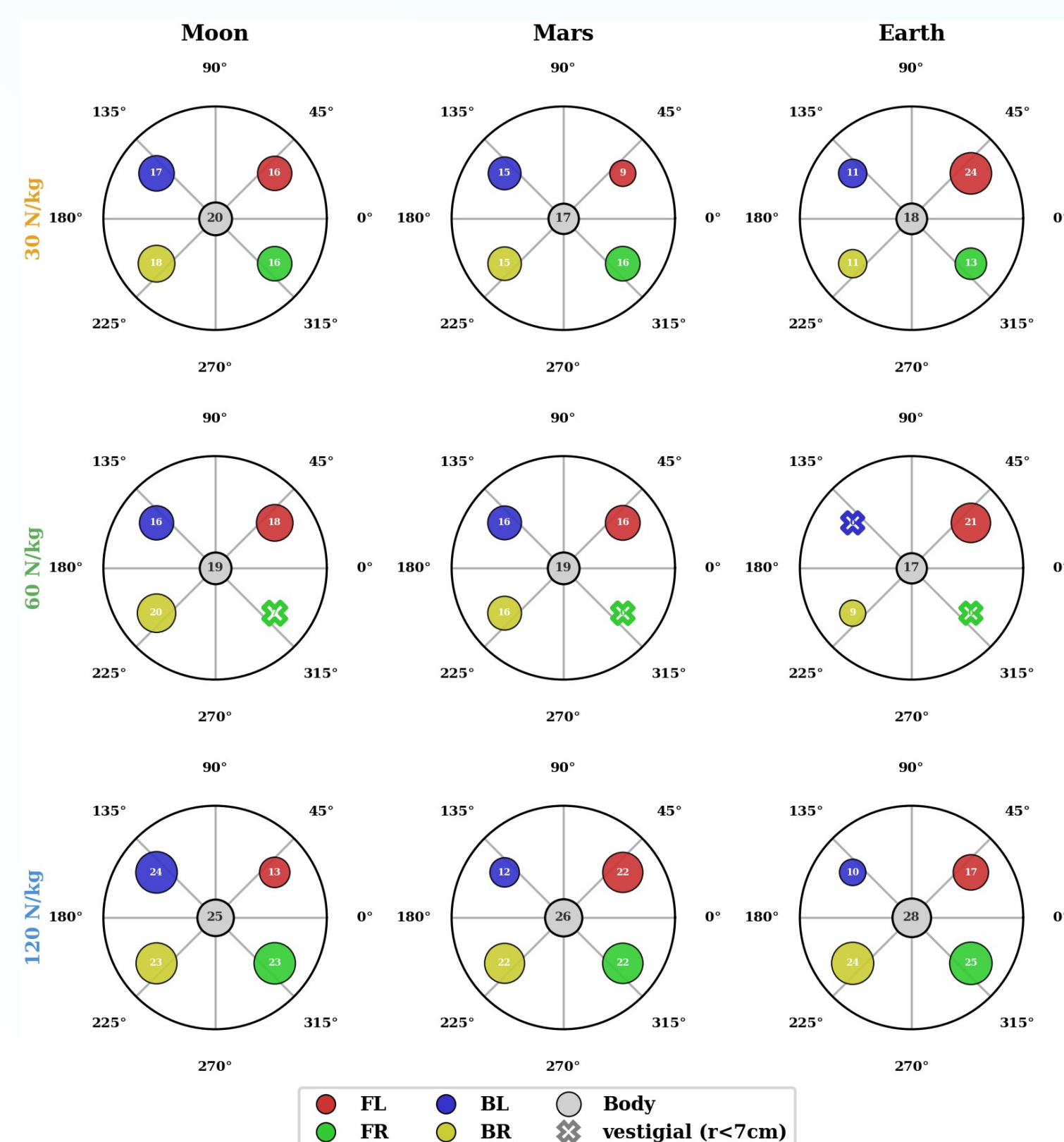
We co-optimize quadruped morphology and control end-to-end through a differentiable physics simulator, with gravity as a first-class design variable. Under a fixed mass budget, we expose regime transitions in body-leg mass allocation, limb morphology, and gait coordination. The framework reveals destination-specific design trends for mass-limited planetary robots and shows that morphology gradients are comparable to control gradients.

Introduction

Once a space mission is selected, the robot's design envelope is fixed and hardware cannot change after launch. A robot on Titan operates in a fundamentally different regime than one on Venus: ballistic timescales, stance loading, and stability margins all shift in ways that reorganize which morphologies and gait patterns are feasible. We sweep gravity not to design a universal robot, but to expose regime transitions and derive destination-specific design guidance under a unified co-optimization pipeline. Engineering practice typically decouples morphology and control: body first, controller second – missing synergistic solutions that only emerge from joint optimization.

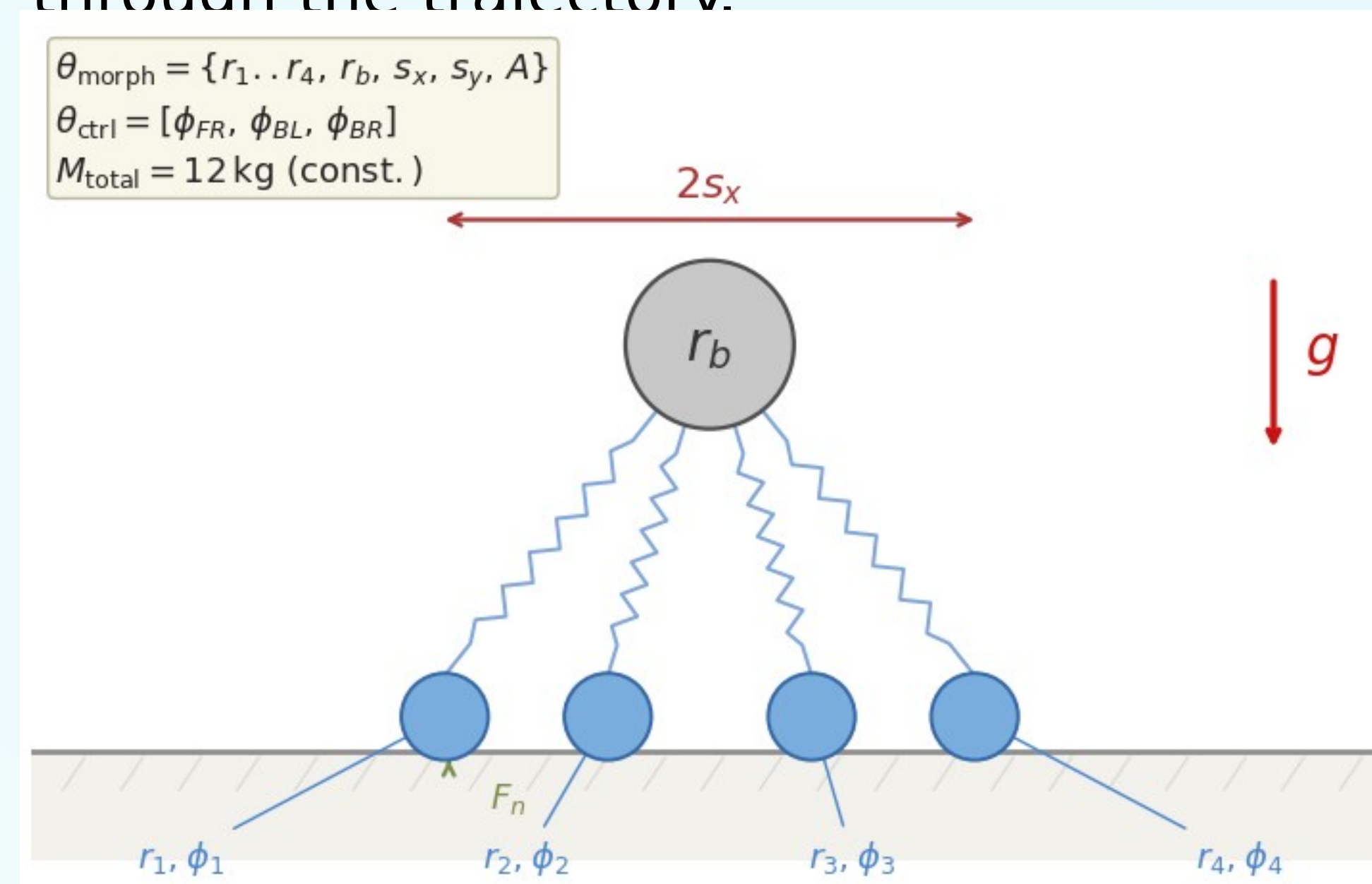
Materials

Gait coordination shifts from aerial flight phases under low gravity to ground-bound walking under high gravity. At 30 N/kg aerial patterns account for 65 % of Moon survivors, declining to 0 % at Earth where walk coordination takes over entirely. At 120 N/kg, aerial gaits persist above 24 % even at Earth: stronger actuators retain the ability to generate ground-clearance forces.



Methodology

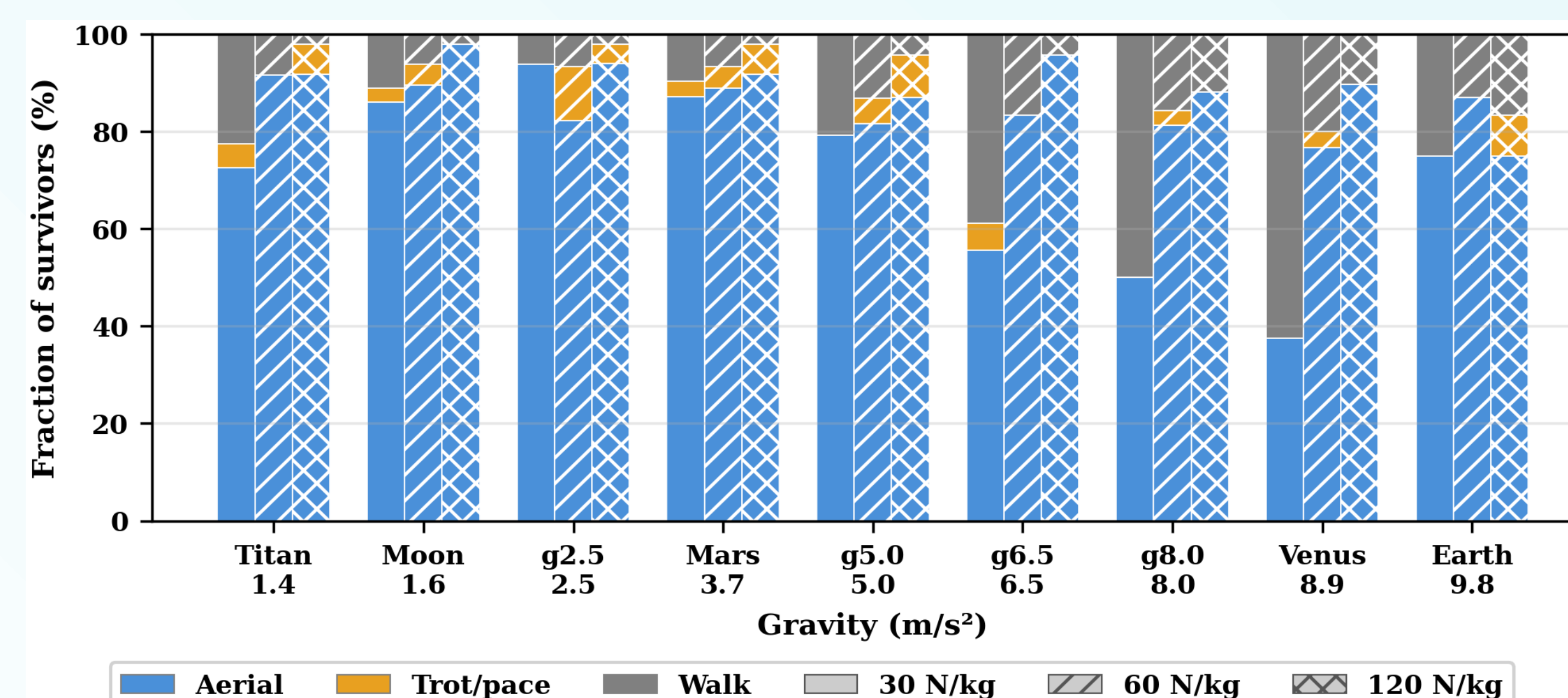
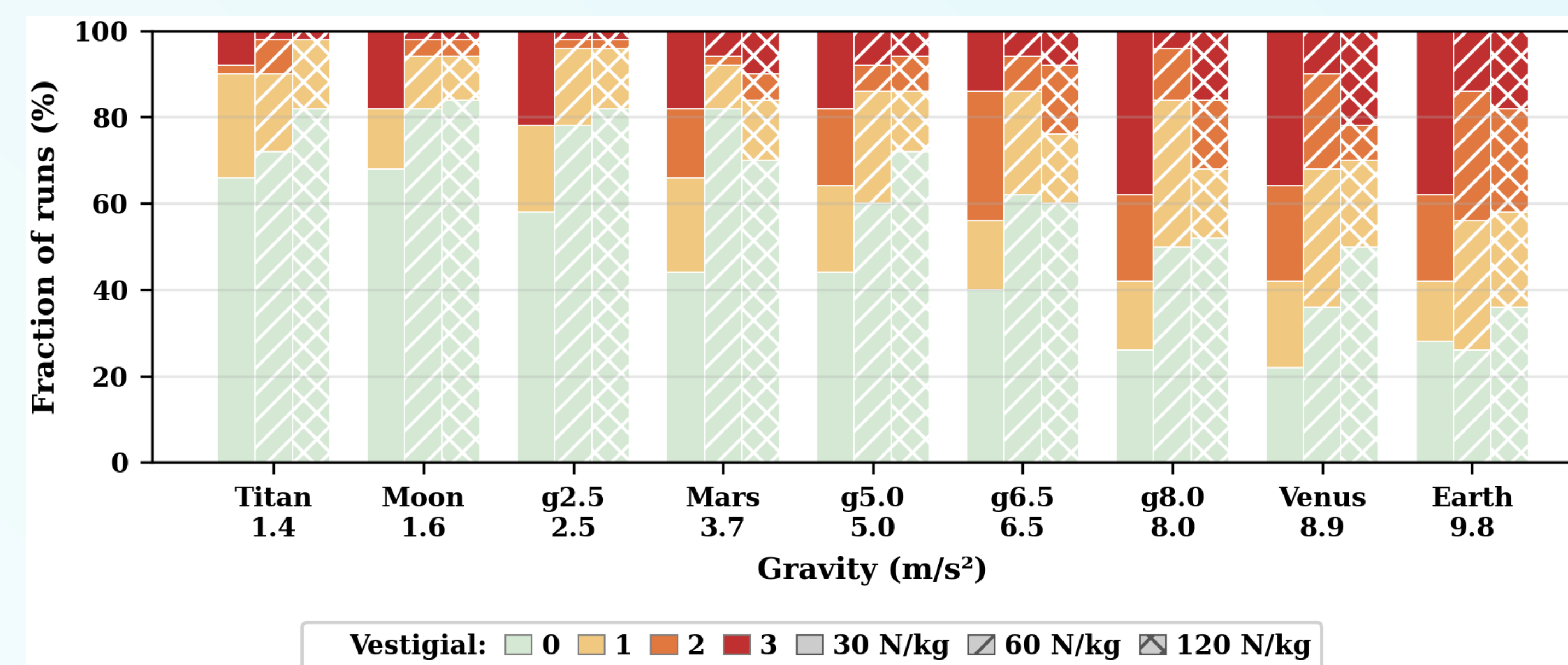
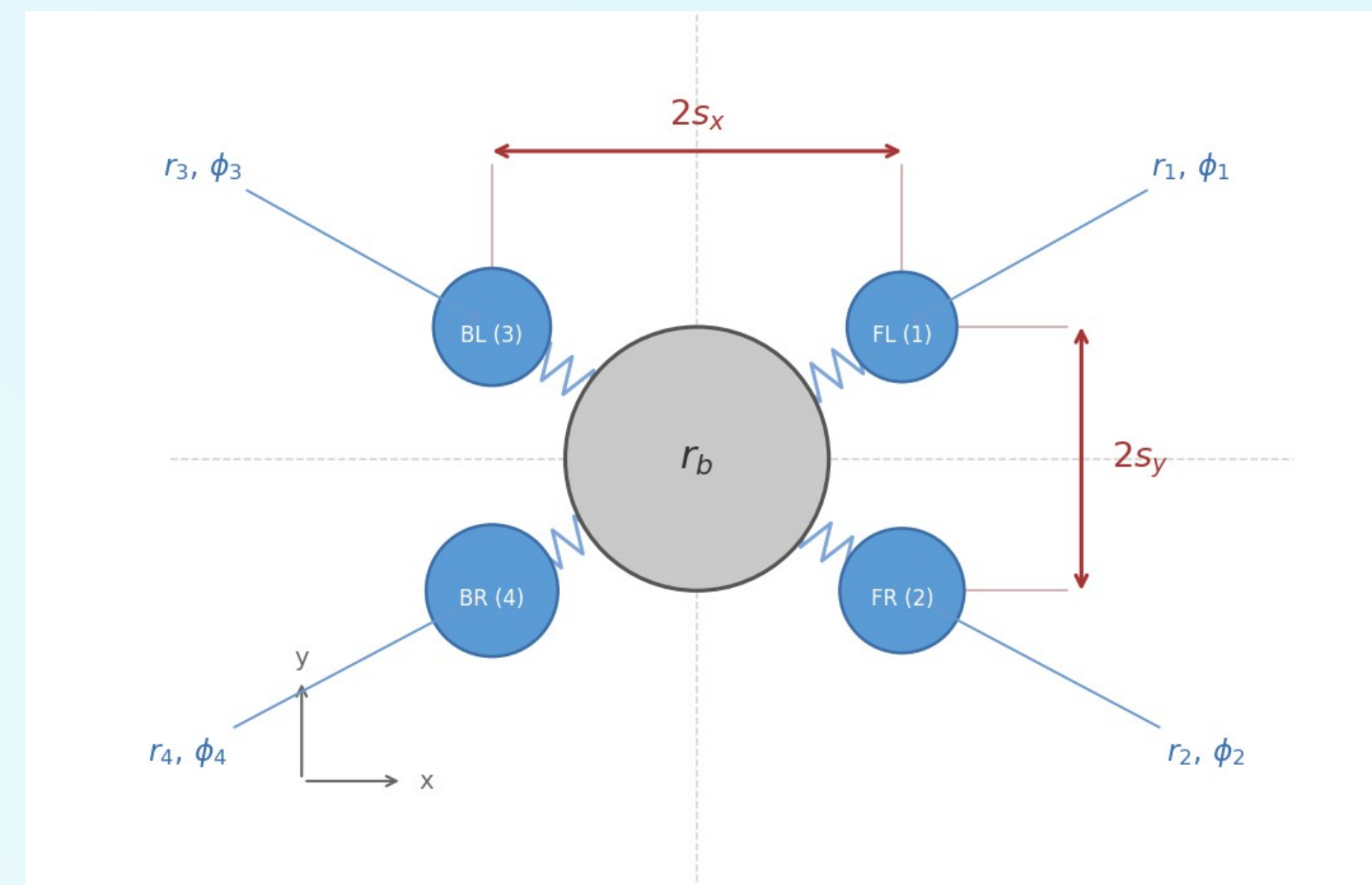
Robot model: five-sphere quadruped, total mass $M = 12$ kg, Growing the body shrinks the legs and vice versa. Locomotion: CPG-driven elliptical foot paths on a 25° slope. PD actuators with mass-proportional caps $a_{\max} \in \{30, 60, 120\}$ N/kg. Simulator: NVIDIA Warp, 2000-step rigid-body contact dynamics, reverse-mode AD through the trajectory.



11-D design vector θ :
Morphology (8): $r_1 \dots r_4, r_b, s_x, s_y, A$
Control (3): $\phi_{FR}, \phi_{BL}, \phi_{BR}$

Loss: $\mathcal{L} = -d_x + \lambda_h (\bar{z} - z)^2 + \lambda_g \langle \vartheta_{b2} \rangle + \lambda_v \langle v_{z2} \rangle + \lambda_E \langle \tau^2 \rangle^*$

Sweep: 9 gravity levels $\times$ 3 force caps, 20–50 seeds per condition, 900 total trials.



Results

Body mass fraction falls monotonically with gravity. At 60 N/kg it drops from 84 % on Titan to 50 % on Earth: mass migrates outward to grow legs and raise per-leg force capacity. At 120 N/kg the shallower range (84 % $\rightarrow$ 67 %) reflects that stronger actuators reduce the imperative to grow legs. Under high gravity and tight force caps, some legs shrink below 7 cm and become dynamically inactive: the optimizer concentrates mass into the remaining limbs. The plateau appears near 8 m/s². Contact-load statistics confirm vestigial legs contribute negligible ground reaction force.

Conclusion

Morphology gradients are comparable to and often exceed control gradients across every gravity level: joint optimization is not redundant.

Key findings:

- Body mass fraction tracks gravity
- Vestigial legs emerge at high g
- Aerial $\rightarrow$ walk transition with gravity
- Morphology gradients $\geq$ control gradients

Future work: bridge five-sphere abstractions to articulated kinematic chains with realistic joint limits and motor torque curves.

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